

# Gold Future/ETF Analysis

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Table 1: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

| <b>Maturity</b> | <b>Slope</b> | <b>Intercept</b>         | <b><math>R^2</math></b> | <b>RMSE</b> |
|-----------------|--------------|--------------------------|-------------------------|-------------|
| 1-Month         | 0.98218      | $-9.0480 \times 10^{-5}$ | 0.92459                 | 0.00269     |
| 2-Month         | 0.98709      | $-1.2547 \times 10^{-4}$ | 0.92244                 | 0.00275     |
| 6-Month         | 0.98870      | $-9.0354 \times 10^{-5}$ | 0.91718                 | 0.00285     |
| 12-Month        | 0.92794      | $9.3489 \times 10^{-5}$  | 0.76020                 | 0.00520     |

Table 2: A summary of the slopes and  $R^2$  from the regressions of futures returns versus gold returns over different holding periods.

|              | <b>Days</b> | <b>1-Month</b> | <b>2-Month</b> | <b>6-Month</b> | <b>12-Month</b> |
|--------------|-------------|----------------|----------------|----------------|-----------------|
| <b>Slope</b> | 1           | 0.96061        | 0.96608        | 0.96757        | 0.91176         |
|              | 5           | 0.99723        | 1.00073        | 0.99971        | 0.94958         |
|              | 10          | 0.99556        | 0.99782        | 0.99821        | 0.97219         |
|              | 15          | 0.99821        | 1.00003        | 0.99906        | 0.98087         |
|              |             |                |                |                |                 |
| $R^2$        | 1           | 0.87699        | 0.87470        | 0.87272        | 0.73603         |
|              | 5           | 0.96413        | 0.96345        | 0.96008        | 0.87631         |
|              | 10          | 0.98020        | 0.97931        | 0.97586        | 0.93513         |
|              | 15          | 0.98603        | 0.98532        | 0.98191        | 0.95316         |
|              |             |                |                |                |                 |

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of \$1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is and \$8.23 for the out of sample period.

Table 3: A summary of the weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

| <b>Portfolio</b>    | $w_0$   | $w_1$   | $w_2$    | $w_3$    | $w_4$   | <b>RMSE<sub>in</sub></b> | <b>RMSE<sub>out</sub></b> |
|---------------------|---------|---------|----------|----------|---------|--------------------------|---------------------------|
| 1-m                 | 0.00263 | 0.99735 |          |          |         | \$2.83                   | \$5.13                    |
| 2-m                 | 0.00687 | 0.99313 |          |          |         | \$3.84                   | \$5.72                    |
| 6-m                 | 0.02215 | 0.97785 |          |          |         | \$10.10                  | \$13.41                   |
| 12-m                | 0.04672 | 0.95328 |          |          |         | \$19.27                  | \$22.90                   |
| 1-m, 2-m            | 0.00165 | 1.23652 | -0.23817 |          |         | \$2.78                   | \$5.04                    |
| 1-m, 6-m            | 0.00169 | 1.04633 | -0.04804 |          |         | \$2.79                   | \$4.87                    |
| 1-m, 12-m           | 0.00200 | 1.01343 | -0.01544 |          |         | \$2.81                   | \$5.02                    |
| 2-m, 6-m            | 0.00350 | 1.22553 | -0.22903 |          |         | \$3.39                   | \$4.65                    |
| 2-m, 12-m           | 0.00342 | 1.08973 | -0.09314 |          |         | \$3.46                   | \$5.13                    |
| 6-m, 12-m           | 0.00614 | 1.68583 | -0.69198 |          |         | \$6.64                   | \$14.41                   |
| 1-m, 2-m, 6-m       | 0.00156 | 1.20157 | -0.18857 | -0.01457 |         | \$2.78                   | \$4.98                    |
| 1-m, 2-m, 12-m      | 0.00174 | 1.27744 | -0.28654 | 0.00734  |         | \$2.78                   | \$5.08                    |
| 1-m, 6-m, 12-m      | 0.00204 | 1.10142 | -0.16131 | 0.05785  |         | \$2.76                   | \$4.80                    |
| 2-m, 6-m, 12-m      | 0.00349 | 1.22294 | -0.22414 | -0.00229 |         | \$3.39                   | \$4.65                    |
| 1-m, 2-m, 6-m, 12-m | 0.00189 | 1.24562 | -0.17704 | -0.12664 | 0.05617 | \$2.75                   | \$4.88                    |